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Bill Hrusa

Math Dept. CMU

Cell 412-965-1159

wh15 @ andrew.cmu.edu

"Math Boot Camp"

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Square Roots

What is $\sqrt{9}$?

± 3 (One Possible Answer)

$$x^2 = 9 \iff x = \pm 3$$

$$\sqrt{9} = 3$$

Convention we
adopt

$\sqrt{x} \geq 0$ ($\sqrt{}$ should be a fn)

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$$\sqrt{x^2} = |x|$$

$$\sqrt{(-3)^2} = \sqrt{9} = 3 = |-3|$$

Import Concepts
from 1-Variable Calculus

1. Limits (& Continuity)
2. Idea of Derivative

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3. Basic Rules for Differentiation.
4. Optimization (Max, Min)
5. Chain Rule
6. Mean Value Theorem
7. Taylor's Thm (with Remainder)
8. Integral as a Limit of Sums
9. Antiderivatives, Indef. Integrals
10. Fund. Thm of Calculus

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11. Techniques of Integration
(incl. numerical)
12. Differentiation of Integrals
13. Integrals in Probability
14. Moment Generating Fns

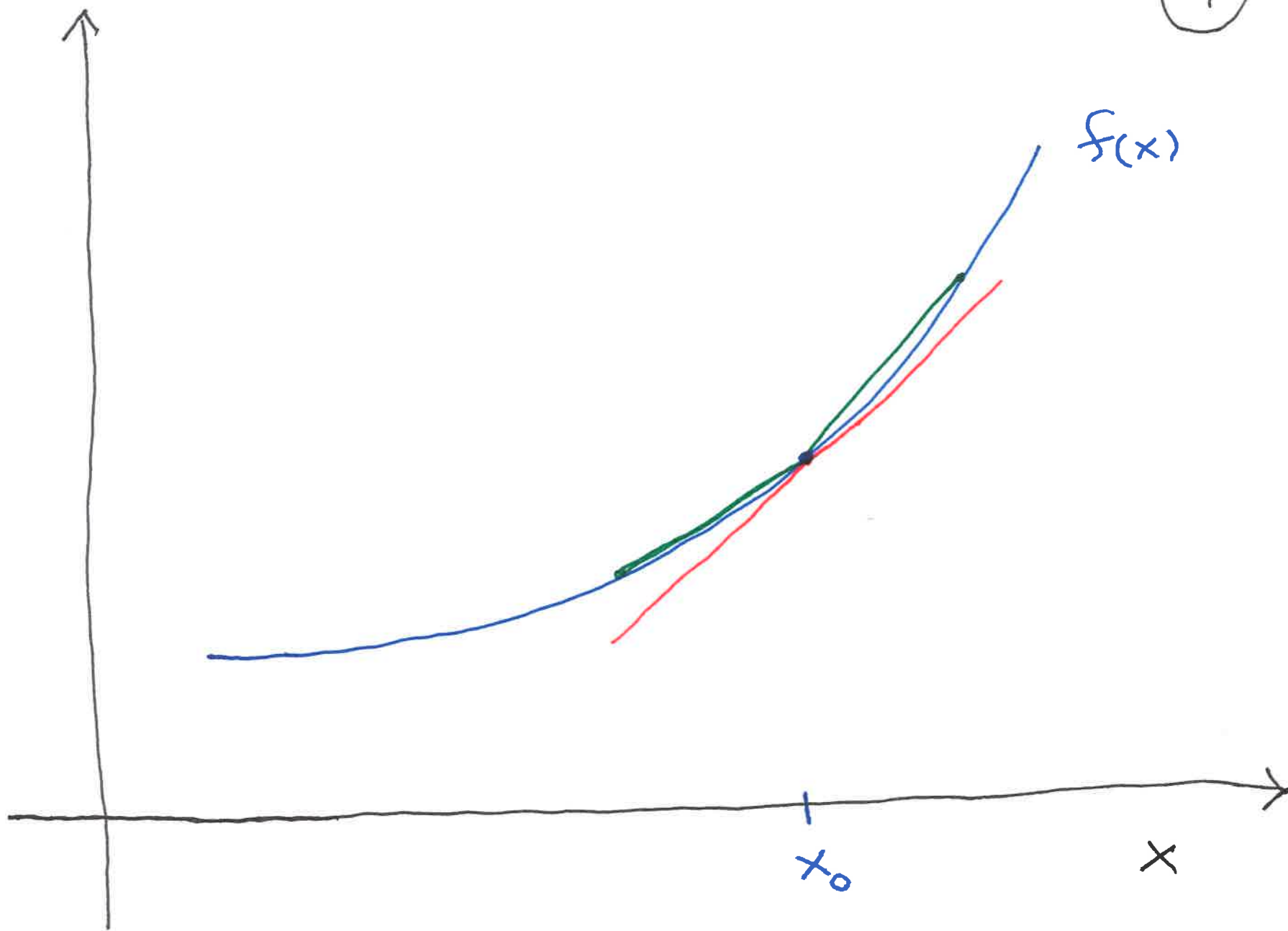
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Derivative of a Function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

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What does differentiability really mean?

Linear Approximation

$$f(x) = \underbrace{f(x_0) + f'(x_0)(x-x_0)}_{\text{linear appx (tangent)}} + \underbrace{e(x_0; x)(x-x_0)}_{\text{error}}$$

where $\lim_{x \rightarrow x_0} e(x_0; x) = 0$

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Other Notations

$$y = f(x)$$

$$f'(x) = \frac{dy}{dx}$$

$$y' = f'(x)$$

$$\frac{d}{dx} f(x) = f'(x)$$

$$y = \sqrt{x}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}, \quad x > 0$$

$$\frac{d}{dx} (\sqrt{x}) = \frac{1}{2\sqrt{x}}, \quad x > 0$$

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Basic Rules

$$\text{I. } \frac{d}{dx}(c_1 f(x) + c_2 g(x)) = c_1 f'(x) + c_2 g'(x)$$

$$\text{II. } \frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

$$\text{III. } \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}$$

$$g(x) \neq 0$$

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Chain Rule: If f and g are diff.

then $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$

$$y = f(u), \quad u = g(x), \quad y = f(g(x))$$

$$x \xrightarrow{g} u \xrightarrow{f} y$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$\frac{dz}{ds} = \frac{dz}{dr} \frac{dr}{ds}$$

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$$1. \frac{d}{dx}(c) = 0$$

$$2. \frac{d}{dx}(x^p) = px^{p-1}$$

$$x > 0, p \neq 0$$

$$3. \frac{d}{dx}(e^x) = e^x$$

$$4. \frac{d}{dx} \ln x = \frac{1}{x}$$

$$\ln x = \log_e x$$

$$5. \frac{d}{dx} \sin x = \cos x, \quad \frac{d}{dx} \cos x = -\sin x$$

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Ex: (a) $f(x) = \sqrt{x^2+1} = (x^2+1)^{1/2}$

$$f'(x) = \frac{1}{2} (x^2+1)^{-1/2} \frac{d}{dx} (x^2+1)$$

$$= \frac{1}{2} (x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}$$

(b) $f(x) = x^2 e^{x^3}$

$$f'(x) = x^2 \frac{d}{dx} (e^{x^3}) + 2x e^{x^3}$$

$$= x^2 (e^{x^3} \cdot 3x^2) + 2x e^{x^3}$$

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What is e ?

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

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Logarithmic Diff

$$y = a^x, \quad a > 0$$

$$\ln y = x (\ln a)$$

$$\frac{d}{dx} (\ln y) = \frac{d}{dx} [x (\ln a)] = \ln a$$

$$\frac{1}{y} \frac{dy}{dx} = \ln a$$

$$\frac{dy}{dx} = y \ln a = a^x (\ln a)$$

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Handy Formula

$$(*) \quad \log_a x = \frac{\ln x}{\ln a}$$

Ex: $y = (x^2 + 1)^{\sin x}$

$$\ln y = \ln[(x^2 + 1)^{\sin x}] = \sin x \ln(x^2 + 1)$$

$$\frac{1}{y} \frac{dy}{dx} = \sin x \left(\frac{2x}{x^2 + 1} \right) + \cos x \ln(x^2 + 1)$$

$$\frac{dy}{dx} = (x^2 + 1)^{\sin x} \left[\frac{2x \sin x}{x^2 + 1} + \cos x \ln(x^2 + 1) \right]$$

Higher Derivatives

$$y = f(x) \quad \frac{dy}{dx} = f'(x)$$

$$\frac{d}{dx}(f'(x)) = f''(x) = y'' = \frac{d^2 y}{dx^2}$$

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = f'''(x) = y'''$$

$$\frac{d^n y}{dx^n} = f^{(n)}(x)$$

$$f^{(17)}(x)$$

17th deriv.

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Taylor's Theorem: Let $n \geq 0$ be an integer. Assume f is $(n+1)$ -times diff on an open interval I containing x_0 . Let $x \in I$ be given. Then

$$\begin{aligned} f(x) = & f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2}f''(x_0)(x-x_0)^2 \\ & + \frac{1}{3!}f'''(x_0)(x-x_0)^3 + \dots \\ & + \frac{1}{n!}f^{(n)}(x_0)(x-x_0)^n \\ & + R_n(x_0; x) \end{aligned}$$

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where

$$R_n(x_0; x) = \frac{1}{(n+1)!} f^{(n+1)}(c) (x-x_0)^{n+1}$$

for some c between x_0, x

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$$

$$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$$

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Conventions

$$0! = 1, \quad (x-x_0)^0 = 1, \quad f^{(0)}(x_0) = f(x_0)$$

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)(x-x_0)^k}{k!}$$

Taylor Polynomial
order n

$$+ \frac{f^{(n+1)}(c)(x-x_0)^{n+1}}{(n+1)!}$$

Remainder

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$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

Taylor Series

[Some "Pathological" fns for
which series converges, but
= $f(x)$ only at $x=x_0$]

$x_0 = 0$ MacLaurin Series

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Important MacLaurin Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots \quad |x| < 1$$

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Ex: $g(x) = e^{x^2}$

Let's Compute $g^{(38)}(0)$

$$g'(x) = e^{x^2} \cdot 2x \quad g''(x) = e^{x^2} \cdot 2 + e^{x^2} (2x)^2$$

$$g'''(x) = \text{mess}$$

Quick Way

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$g(x) = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots + \frac{x^{2k}}{k!} + \dots$$

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$$g(x) = g(0) + g'(0)x + \frac{1}{2}g''(0)x^2 + \dots \\ + g^{(n)}(0)\frac{x^n}{n!} + \dots$$

$$1 + x^2 + \frac{x^4}{2!} + \dots + \frac{x^{38}}{19!} + \dots$$

$$= g(0) + g'(0)x + \frac{1}{2}g''(0)x^2 \\ + g^{(38)}(0)\frac{x^{38}}{38!} + \dots$$

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$$\frac{g^{(38)}(0)}{38!} x^{38} = \frac{x^{38}}{19!}$$

$$g^{(38)}(0) = \frac{38!}{19!}$$

$$g^{(2n)}(0) = \frac{(2n)!}{n!}$$

$$g^{(2n+1)}(0) = 0$$